

Methods of Separation of Homogeneous mixtures

Question:-What is a crystal?

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**Answer:-** Crystal is a solid that is made up of particles arranged in a regular geometric pattern.

**Question:-What is crystallization?**

**Answer:-** The process of forming of crystals from a saturated solution is called crystallization.

**Question:- For what purpose crystallization method is used?**

**Answer:-** Crystallization method is used to purify the solids.

**Question:- Principle of crystallization?**

**Answer:-** The principle of purification by crystallization is based on the differences in the solubility of a substance at different temperatures.

**Activity:-** Prepare a sample of pure copper sulphate by the method of crystallization.

1. Collect a sample of copper sulphate blue vitriol.

2. Take one gram of copper sulphate and place it in a 100 ml beaker. Add 25 ml of water to the beaker and add a drop of dilute sulphuric acid.
3. Gently heat the mixture in a water bath while stirring constantly.
4. Gradually add more copper sulphate until the solution becomes saturated.
5. Filter the hot solution to remove insoluble impurities. Collect it in a clean beaker and cover it with the watch glass.
6. Allow the solution to cool slowly without disturbing it.
7. It gives enough time to the particles in the solution to come together resulting in the formation of larger shiny well shaped and blue coloured crystals.
8. Filter the crystals and wash them with cold water and allow them to dry on a watch glass.

**Question:- Why sulfuric acid is added in the sample of copper sulphate solution?**

**Answer:-** Sulfuric acid helps in making pure crystals by preventing unwanted reactions.

**Question:- if a hot saturated solution copper sulphate is**

**cooled rapidly in ice cold water smaller and less well formed crystals will form why?**

Answer:- With slow cooling, the particles deposit steadily onto the few existing seeds, creating large, perfectly geometric (well-formed) crystals. In ice water, there are so many seeds that the available material is quickly divided among them, leaving insufficient building blocks for large crystals.

**Question:- What is distillation?**

Answer:- A homogeneous mixture of two miscible liquids can be separated by heating the mixture until the liquid with the lower boiling point vaporises. The vapour is then cooled turning back into liquid. This process is known as distillation.

**Question:- Necessary condition required for distillation?**

Answer:- It allows the recovery of the solvent or the separation of liquids that differ in boiling point by at least about 25 degree Celsius.

**Question:- Method of distillation.**

Answer During the process vapours of the lower boiling



liquid are pass through a condenser where they are cooled usually by circulating water or air and condensed as a pure liquid. The pure liquid is collected in a separate vessel, while the solid or other liquid from the mixture remains in the distillation flask.

**Question:-How the mixture of acetone and water be separated?**

**Answer:-**

A mixture of acetone and water can be separated by distillation because their boiling points differ sufficiently. Acetone boils at about 56 degree Celsius and water boils at 100 degree Celsius. A large difference in boiling points allows one liquid to vaporize before the vapour vapours of the other liquid.

**Question -What is fractional distillation?**

**Answer:-** Fraction distillation is the process of separating components of a mixture with the relatively small differences less than 25 degree celsius in their boiling points.

Crude petroleum is extracted from the Earth's crust and separated into these different fractions using the

process of fractional distillation.

**Question -What is paper chromatography?**

**Answer** The method of separation of the components based on the colour on chromatography paper is known as paper chromatography.

**Activity:- Separate out different colours in blue ink.**

**Solution:-**

1. Take a 3 cm wide strip of chromatography paper and draw a straight horizontal line 2 cm from the bottom of the paper with the pencil.
2. Mark a spot with a black sketch pen at the centre of the line.
3. Take enough water to make a thin layer at the bottom of a glass jar or a cylinder.
4. Place the paper strip with the ink spot vertically into the container so that it's lower end dips into the water. The water level should be below the spots.
5. As the solvent rises the ink starts to separate into different colour spots.

**Question -What is the principle of chromatography?**

Answer:-

It uses differences in the interaction of the components with the solvent and the paper to separate them. The liquid carries the substances up the paper separating them based on how fast they move.

**Note :-** You can separate different pigments present in the green extract of **spinach leaves** or coloured pigments from the **flower petals**.

**Question** Difference between evaporation and crystallization?

Answer:-

Evaporation

1. To separate a solid from its liquid solution completely.
2. The solvent is continuously heated until it fully boils off.
3. Soluble impurities remain trapped in the dry residue.

Crystallization

1. To separate and purify a solid substance.
2. The solution is carefully heated to become saturated, then slowly cooled so crystals form.



3. Impurities remain dissolved in the liquid while pure crystals grow.

**Question:-Which method is to choose?**

**Answer:-**

Use Evaporation if you are separating a heat-stable substance like salt in water.

Use Crystallization if the substance is delicate, degrades with high heat, or requires a high degree of purity.

**Method of separation of Heterogeneous mixtures.**

**Question What are immiscible liquids?**

**Answer** The liquids which do not mix with each other are called immiscible liquids.

**Activity:- Separate a mixture of mustard oil from water.**

**Answer**

1. Pour a mixture of 5 ml mustard oil and 20 ml water into a 50 ml separating funnel.
2. Let it stand undisturbed.
3. You will see the formation of two separate layers of mustard oil and water.

4. The yellow coloured mustered oil forms the upper layer and water forms the lower layer.
5. Open the stop Cock of the separating funnel slowly to collect the lower layer of water carefully into a container.
6. Close the stopcock when the water is almost fully drained.
7. Collect the layer of oil separately by opening the stop cock again.

### **Question -What is Sublimation?**

Answer - Substances on heating below its melting point changes directly from the solid state to the vapour state without passing through the liquid state this process is called sublimation.

### **Question -What is deposition?**

Answer - On cooling vapours condense back into a solid without becoming a liquid this is called deposition.

### **Question -What is dry ice?**

Answer -Solid Carbon dioxide is called dry ice.

### **Question -What are alloys?**

Answer -A homogeneous mixture of two or more metals



or a metal and a non metal is called an alloy.

### **Question - Properties of Alloys?**

Answer - 1. Physical methods cannot separate the components of an alloy.

2. Alloys are prepared to produce materials that are stronger and more rigid.

3. Alloys are prepared to produce materials which are more corrosion resistant.

### **Question - What are the constituent of following Alloys**

**Answer**

**Brass.** Copper, Zinc

**Bronze.** Copper, Tin

**Stainless steel** Iron, carbon

chromium, nickel, molybdenum

**Activity:- Separate a mixture of Camphor and sand.**

**Answer:-**

1. Take a spatula full of the mixture containing crushed camphor and sand. Put it into a clean and dry China dish. Place it on the tripod stand using a wire gauze.

2. Take a clean and dry glass funnel. Plug its nozzle with cotton.
3. Keep this funnel inverted on the China dish.
4. Light the burner and place it under the wire gauze.
5. Heat the China for a few minutes.
6. Observe the inner wall of the funnel carefully.
7. You may find white solid camphor deposits on the inner wall of the funnel while sand remains in the China dish.

**Question:- What is centrifugation?**

**Answer:-** Centrifugation is a mechanical process that uses rapid spinning to separate particles in a liquid mixture based on their size, shape, and density.

**Question:- What is the principle of centrifugation?**

**Answer:-** The principle of centrifugation is that denser particles are forced to the bottom and the lighter particles stay at the top when spun rapidly.

**Question:- Examples of centrifugation**

**Answer:-**

1. Centrifugation is used to separate the components of blood.

2. It is used to separate the cream from milk.

**Question:-** what is coagulation?

**Answer:-** It is a process in which alum is added to a beaker containing muddy water. The alum causes the fine suspended particles to clump together.

These large clumps then settle by sedimentation and can be separated from water by filtration.